Predicting Coronary Heart Disease from the BRFSS 2015 Survey with Gradient Boosting and Neural Networks in NumPy

Zhihao Chen, Zhiqian Zhou, Guanyu Chen
School of Computer and Communication Sciences, EPFL, Switzerland

***Abstract*—We predict coronary heart disease (`_MICHD`) from 321 answers of the BRFSS 2015 health survey. The task is imbalanced (8.8% positives) and the data have structured missing values and codebook special codes. We clean the data column by column, compare the linear methods of the course with a gradient boosted tree model (GBM) and a small neural network (MLP), and average the last two. The ensemble reaches a cross-validated F1 of 0.437, 0.434 on a held-out set and 0.447 on AIcrowd (87.7% accuracy). Seven survey questions asked after a cardiac event raise F1 to 0.457 (88.3% accuracy). We exclude them from the official model, because they are not available when screening undiagnosed people.**

I. Introduction

The Behavioral Risk Factor Surveillance System (BRFSS) is a yearly US telephone survey on health and behaviour [1]. The label `_MICHD` states whether a respondent reported coronary heart disease or a myocardial infarction. The data have 328,135 training rows, 109,379 test rows and 321 features. Only 8.83% of the labels are positive. Predicting -1 for everyone gives 91.2% accuracy but F1 $= 0$, so we optimize the F1 of the positive class. Only NumPy is allowed. Our contributions are (i) codebook-driven cleaning, (ii) a NumPy GBM and MLP ensemble, and (iii) an analysis of leaky survey questions [2].

II. Data and preprocessing

Special codes. The 2015 codebook [3] defines special codes per column. Codes 7/9, 77/99 and 777/999 ("don't know", "refused") become missing; 88 and 888 ("none") become 0. The rule is set per column, because 7 or 99 are valid values in some columns. Weight and height are converted to kg and cm, and frequency codes to one unit.

Missing values. 147 of the 321 columns are more than 50% missing. Most gaps follow the survey skip logic and are not random. Eleven skip-logic fills set the implied answer, e.g. "takes blood pressure medication" = no when the person has no high blood pressure. Missingness is informative: the positive rate is 3.8% when `BPMEDS` is missing and 16.3% when it is observed. We therefore add missing-value indicators and one "don't know" flag per question.

Column selection. We drop survey administration fields, columns more than 95% empty and redundant recodes. The cleaned matrix keeps 75 raw and 2 derived features plus 61 "don't know" flags (138 columns).

Fitted pipeline. Median or mode imputation, one-hot encoding of nominal columns (rare levels pooled), $\log(1+x)$ of counts, clinical interactions and standardization give about 140 columns. All statistics are fitted on the training folds only.

III. Models and validation

Validation protocol. We hold out a stratified 20% of the training rows (65,627 rows, seed 2026). All choices use 5-fold stratified cross-validation (CV) on the remaining 262,508 development rows. Preprocessing, the state encoding, the models and the decision rule are fitted inside each fold. A change is accepted only if mean F1 rises by at least 0.001 and it wins on at least 4 of 5 folds. No choice uses AIcrowd or the test set.

Linear baselines. We train the six course methods of `implementations.py` on the fitted pipeline with 10 pairwise interactions. Labels are ± 1 for squared loss and $\{0, 1\}$ for logistic loss. With the default 0.5 threshold, the logistic model reaches F1 0.23; tuning the decision threshold jumps to F1 0.430. Every linear model requires a tuned decision rule.

Gradient boosted trees. Our NumPy GBM minimizes logistic loss with second-order (Newton) leaf values $-G/(H+\lambda)$. Features are binned into 64 histogram bins. Missing values form their own bin, with optimal split directions learned automatically. Trees have depth 5, learning rate 0.05, row and column subsampling 0.7 and 0.8, at least 200 rows per leaf and $\lambda = 5$. Early stopping uses an inner 20% split. The state of residence is replaced by its smoothed prevalence, fitted on the training folds (+0.0026 F1, 4/5 folds).

Neural network. The MLP has one hidden layer of 64 ReLU units. It is trained with Adam (learning rate 10^{-3}, batch 512), weight decay 10^{-3} and early stopping. We average the logits of 10 seeds.

Ensemble and decision rule. The final prediction score is $0.5\, s_{\text{GBM}} + 0.5\, s_{\text{MLP}}$ (logits). We predict $+1$ for the top 14% of respondents. The share is more robust than a fixed threshold under retraining. Fig. 1 shows that F1 remains flat between 12% and 16%.

IV. Results

Table I compares all models on the same folds. Squared-loss methods reach 0.422. The logistic loss adds 0.008. Both

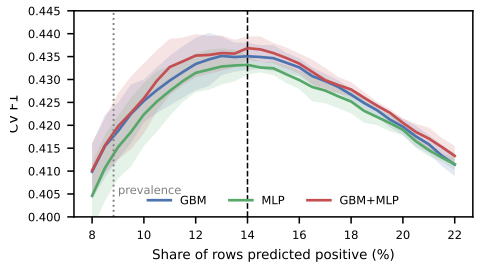


Figure 1. CV F1 (mean ± std over 5 folds) as a function of the share of rows predicted positive. Dotted: prevalence (8.83%). Dashed: our choice (14%).

Table I
F1 OF THE POSITIVE CLASS. CV: 5-FOLD ON THE DEVELOPMENT ROWS, MEAN ± STD, TOP 14% PREDICTED POSITIVE.

Model	CV F1	Held-out
Always −1	0.000	0.000
Least squares / Ridge / MSE GD	0.4216 ± 0.0034	–
MSE SGD (not converged)	0.3853 ± 0.0085	–
Logistic regression (GD)	0.4302 ± 0.0047	–
Reg. logistic regression	0.4300 ± 0.0044	0.4277
GBM (3 seeds)	0.4354 ± 0.0032	0.4343
MLP (10 seeds)	0.4334 ± 0.0026	0.4310
GBM + MLP (final clean)	$\mathbf{0.4368 \pm 0.0026}$	**0.4340**

non-linear models beat the linear baselines, and their average is best. The out-of-fold scores of the GBM and the MLP are correlated at 0.976, which limits the gain of the ensemble.

On the held-out rows the clean model has precision 0.354 and recall 0.561. On AIcrowd it scores F1 0.447 and accuracy 87.7%. The gap with CV is about two bootstrap standard deviations of the public F1 (sd 0.005).

Rejected alternatives. None of the following passed the acceptance rule: SMOTE resampling (F1 fell to 0.285); 2-layer MLPs (0.4272); class weights (-0.003) and focal loss (-0.021); rank normalization (-0.003); PCA (-0.015); feature selection; composite features (-0.0008); crossed target encodings (-0.0010); refitting GBM on 100% rows (-0.0010); Naive Bayes (-0.016), random forest and k-NN; stacking; and per-group shares (-0.001).

Important features. The largest drops in AUC when a feature is permuted come from age (0.035), general health (0.021), sex (0.018), high cholesterol (0.009) and prior stroke (0.008). Simple rules agree: a prior stroke has a lift of 4.24, and five or more chronic conditions a lift of 3.72.

V. LEAKY QUESTIONS

Seven questions are asked after a cardiac event or concern its treatment: cardiac and stroke rehabilitation (`HAREHAB1`, `STREHAB1`), and aspirin use and its reasons (`CVDASPRN`, `RDUCHART`, `RDUCSTRK`, `RLIVPAIN`, `ASPUNSAF`). Their strength differs. All 615 development respondents who answered `HAREHAB1` are positive (158 of 158 in the held-out set). `STREHAB1` has a positive rate of 36%, the other five 5–21%. Such features are a form of target leakage [2].

Table II
CLEAN (OFFICIAL) MODEL VS. THE LEAKY VARIANT. ABLATION: NESTED CV F1, CLEAN → LEAKY.

Model	CV F1	Held-out
Clean (top 14%)	0.4368 ± 0.0026	0.4340
Leaky (top 13%)	0.4464 ± 0.0032	0.4445
Ridge	$0.4210 \rightarrow 0.4297$ (+0.009)	
Reg. logistic	$0.4294 \rightarrow 0.4414$ (+0.012)	
GBM	$0.4330 \rightarrow 0.4444$ (+0.011)	
MLP	$0.4326 \rightarrow 0.4435$ (+0.011)	
GBM + MLP	$0.4366 \rightarrow 0.4452$ (+0.009)	

Table II shows that adding them improves every model by 0.009 to 0.012, on 5 of 5 folds. `HAREHAB1` alone gives +0.008, and the aspirin items +0.004 more. On AIcrowd, the leaky ensemble scores F1 0.457 and accuracy 88.3%.

The gain is real for this benchmark, but it depends on the context of use. A screening model is meant for people without a diagnosis. These people have never been offered cardiac rehabilitation, so the answer that drives the gain does not exist for them. We therefore keep the clean model as our official submission and report the leaky variant separately (`run.py -with_leakage`).

VI. DISCUSSION

All model families reach a similar F1 of 0.42–0.44, suggesting a ceiling for self-reported survey data. Most errors are "silent" cases: among false negatives, 56% are women, 40% lack high blood pressure, 36% report excellent or very good health, and 96% report no prior stroke. These respondents provide little signal. Higher recall would require clinical measurements such as blood pressure, lipids, or an ECG. The public leaderboard is also noisy: a 0.5-point shift in test prevalence moves F1 by roughly 0.01.

VII. SUMMARY

A NumPy ensemble of gradient boosted trees and a neural network, combined with codebook cleaning, reaches F1 0.434 on held-out data and 0.447 on AIcrowd (87.7% accuracy). Survey questions asked after a cardiac event add about 0.01 F1, reaching 0.457 on AIcrowd (88.3% accuracy). We exclude them from screening, as they are unavailable for undiagnosed individuals.

REFERENCES

[1] Centers for Disease Control and Prevention (CDC), "Behavioral risk factor surveillance system survey data 2015," https://www.cdc.gov/brfss/annual_data/annual_2015.html, 2015.

[2] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, "Leakage in data mining: Formulation, detection, and avoidance," *ACM Transactions on Knowledge Discovery from Data*, vol. 6, no. 4, pp. 15:1–15:21, 2012.

[3] Centers for Disease Control and Prevention (CDC), "LLCP 2015 codebook report, behavioral risk factor surveillance system," https://www.cdc.gov/brfss/annual_data/2015/pdf/codebook15_llcp.pdf, 2016.